

Intermediate Macro

Chapter 11: Saving, Capital, Accumulation, and Output

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11.4 Physical versus Human Capital



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Key Tools, Concepts, and Assumptions

Tools and Concepts

- Chapter 11 develops the **Solow model** of growth for the case of no technological change and no population growth.
- The **golden-rule level of capital** per worker is the value of capital worker that maximizes steady-state consumption per worker.
- The **Cobb-Douglas** production function is described in an appendix.

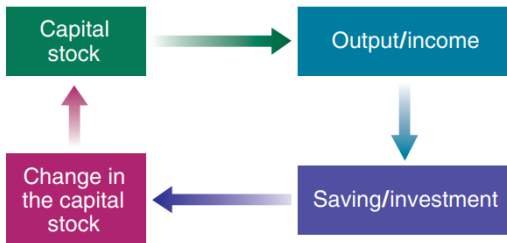
Assumptions

- We assume a closed economy, a fixed labor force, and a fixed level of technology.

Capital, Output, and Saving/Investment

11.1 Interactions between Output and Capital

- There are two relations between output and capital;
 - ① The amount of capital determines the amount of output produced.
 - ② The amount of output determines saving which in turn determines capital accumulation over time.



The Effects of Capital on Output

11.1 Interactions between Output and Capital

- Recall that under the assumption of constant returns to scale, we can write the following relation between output and capital per worker (equation (10.3)):

$$\frac{Y}{N} = F\left(\frac{K}{N}, 1\right)$$

- To save notation, write the aggregate production function as

$$y = \frac{Y}{N} = f\left(\frac{K}{N}\right) = f(k)$$

First Relation between Output and Capital per Worker

11.1 Interactions between Output and Capital

- Now make the following assumptions
 - ① There is no technological change.
 - ② Population, the labor force participation rate, and the natural rate of unemployment are all constant. Thus, N , interpreted as the natural level of employment, is also constant.
- With the introduction of time indexes (t) for output and capital, the production function can now be written as:

$$y_t = \frac{Y_t}{N} = f\left(\frac{K_t}{N}\right) = f(k_t) \quad (11.1)$$

- Now we can see that higher capital per worker leads to higher output per worker.

Output and Investment

11.1 Interactions between Output and Capital

- To derive the relationship between output and investment we make three assumptions.

- ① The economy is closed. Investment, I , is equal to savings (private saving + public saving).

$$I = S + (T - G)$$

- ② Public saving ($T - G$) is equal to zero. This assumption allows us to focus on private saving. So, investment is equal to private saving, or $I = S$.
- ③ Private saving is proportional to income;

$$S = sY$$

- Now we can combine these relations and see the following relationship between investment and output:

$$I_t = sY_t$$

Investment and Capital Accumulation

11.1 Interactions between Output and Capital

- The second step relates investment, which is a flow, to capital, which is a stock.
- Assume that capital depreciates at rate δ .
- The evolution of the capital stock is then given by

$$K_{t+1} = (1 - \delta)K_t + I_t$$

- Investment creates new capital, but the existing capital stock depreciates. Any change in capital at the end of the year is the net effect of new investment and depreciation of existing capital.
- Replacing investment by its expression above and dividing both sides by N gives

$$k_{t+1} = \frac{K_{t+1}}{N} = (1 - \delta)\frac{K_t}{N} + s\frac{Y_t}{N} = (1 - \delta)k_t + sy_t$$

Evolution of Capital per Worker

11.1 Interactions between Output and Capital

- Expanding the term $(1 - \delta)k_t$ to $k_t - \delta k_t$, moving k_t to the left, and reorganizing the right side,

$$k_{t+1} - k_t = sy_t - \delta k_t \quad (11.2)$$

- We have derived two relations:
 - From the production side, we have seen in equation (11.1) how capital determines output.
 - From the saving side, we have seen in equation (11.2) how output in turn determines capital accumulation.
- We can now put the two relations together and see how they determine the behavior of output and capital over time.

Dynamics of Capital and Output

11.2 The Implications of Alternative Saving Rates

- Replacing output per worker y_t in equation (11.2) by its expression in terms of capital worker k_t from equation (11.1) gives

$$k_{t+1} - k_t = sf(k_t) - \delta k_t \quad (11.3)$$

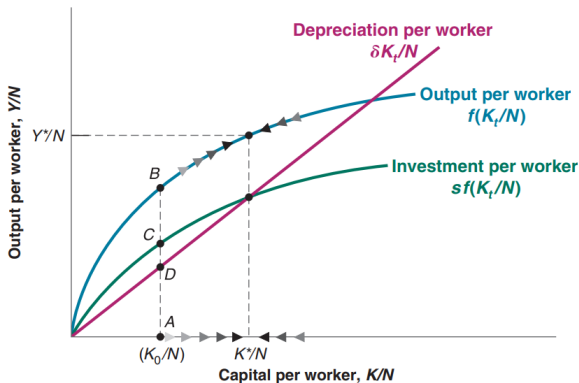
change in capital = investment – depreciation

- Capital per worker increases to the extent that total saving per worker exceeds depreciation of the existing capital stock per worker.

Capital and Output Dynamics

11.2 The Implications of Alternative Saving Rates

- Equations (11.3) and (11.1) contain all the information we need to understand the dynamics of capital and output over time. The easiest way to interpret them is to use a graph.



Long-Run Equilibrium in the Economy

11.2 The Implications of Alternative Saving Rates

- Let us look more closely at the levels of output per worker and capital per worker to which the economy converges in the long run.
- The state in which output per worker and capital per worker are no longer changing is called the **steady state** of the economy.
- Setting the left side of equation (11.3) equal to zero, the steady-state value of capital per worker k^* is given by

$$sf(k^*) = \delta k^* \quad (11.4)$$

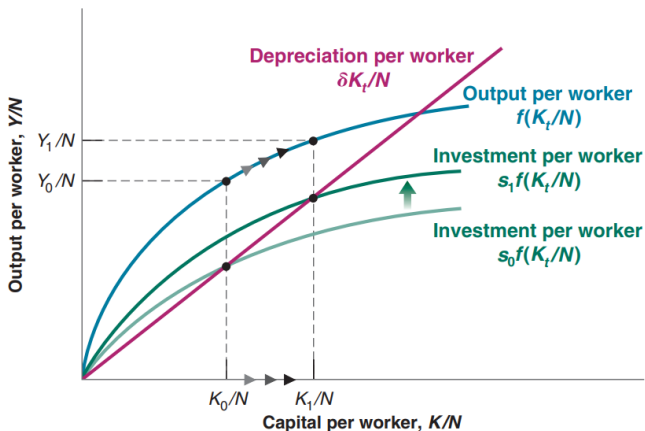
- Given steady-state capital per worker, the steady-state value of output per worker is given by the production function

$$y^* = f(k^*) \quad (11.5)$$

The Saving Rate and Output

11.2 The Implications of Alternative Saving Rates

- Now consider an increase in the saving rate.



The Effects of Different Saving Rates

11.2 The Implications of Alternative Saving Rates

- How does the saving rate affect the growth rate of output per worker?

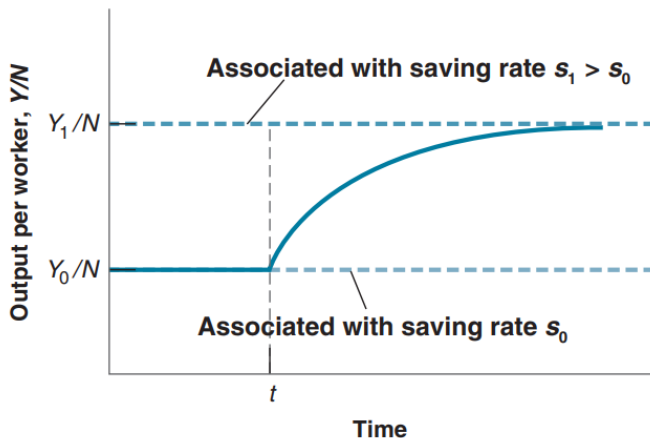
The Effects of Different Saving Rates

11.2 The Implications of Alternative Saving Rates

- How does the saving rate affect the growth rate of output per worker?
- Our analysis leads to a three-part answer:
 - ① The saving rate has *no effect* on the *long-run growth rate* of output per worker, which is equal to zero.
 - ② Nonetheless, the saving rate determines the *level* of output per worker in the long run. Other things being equal, countries with a higher saving rate will achieve higher output per worker in the long run.
 - ③ An increase in the saving rate will lead to higher growth of output per worker for some time, but not forever.

The Effects of an Increase in the Saving Rate (without Technical Progress)

11.2 The Implications of Alternative Saving Rates



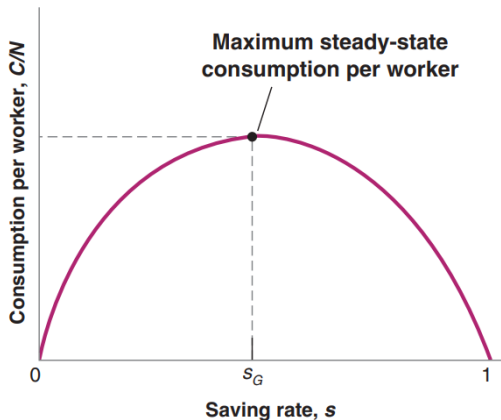
The Saving Rate and Consumption

11.2 The Implications of Alternative Saving Rates

- What is the optimal saving rate?
- A very low saving rate will result in very low steady-state output and consumption per worker. A very high saving rate will waste resources on depreciation, since extra units of capital per worker produce very little extra output per worker when the capital stock is high.
- Somewhere in between is a saving rate that *maximizes steady-state consumption* per worker.
- This rate is called the **golden-rule saving rate**, which produces the **golden-rule level of capital**.

The Effects of the Saving Rate on Steady-State Consumption per Worker

11.2 The Implications of Alternative Saving Rates



A Cobb-Douglas Production Function

11.3 Getting a Sense of Magnitudes

- Assume the production function is

$$Y = \sqrt{K}\sqrt{N} \quad (11.6)$$

- Dividing both sides by N (because we are interested in output per worker)

$$y = \frac{Y}{N} = \frac{\sqrt{K}\sqrt{N}}{N} = \frac{\sqrt{K}}{\sqrt{N}} = \sqrt{\frac{K}{N}} = \sqrt{k}$$

- The production function f relating output per worker to capital per worker is given by

$$f(k_t) = f\left(\frac{K_t}{N}\right) = \sqrt{\frac{K_t}{N}} = \sqrt{k_t}$$

- Replacing $f(k_t)$ by $\sqrt{k_t}$ in equation (11.3),

$$k_{t+1} - k_t = s\sqrt{k_t} - \delta k_t \quad (11.7)$$

Steady-State Capital per Worker

11.3 Getting a Sense of Magnitudes

- Start with equation (11.7). In steady-state the amount of capital per worker is constant, so the left side of the equation is equal to zero. This implies

$$s\sqrt{k^*} = \delta k^*$$

- Square both sides:

$$s^2 k^* = \delta^2 (k^*)^2$$

- Divide both sides by k^* and reorganize:

$$k^* = \left(\frac{s}{\delta}\right)^2 \quad (11.8)$$

The Effects of the Saving Rate on Steady-State Output

11.3 Getting a Sense of Magnitudes

- From equations (11.6) and (11.8), steady-state output per worker is given by

$$y^* = \sqrt{k^*} = \sqrt{\left(\frac{s}{\delta}\right)^2} = \frac{s}{\delta} \quad (11.9)$$

- In this case, if the saving rate doubles, so does long-run output per worker.
- How fast does the economy adjust?

The Dynamic Effects of an Increase in the Saving Rate

11.3 Getting a Sense of Magnitudes

- Suppose that s increases in year 0 from 0.1 to 0.2, that the depreciation rate equals 0.1, and the initially $k_0 = 1$.
- In year 0, *nothing happens to the capital stock* (recall that it takes one year for higher saving and higher investment to show up in higher capital).
- In year 1, equation (11.7) gives

$$k_1 - k_0 = s\sqrt{k_0} - \delta k_0$$

- With a depreciation rate equal to 0.1 and a saving rate now equal to 0.2, this equation implies

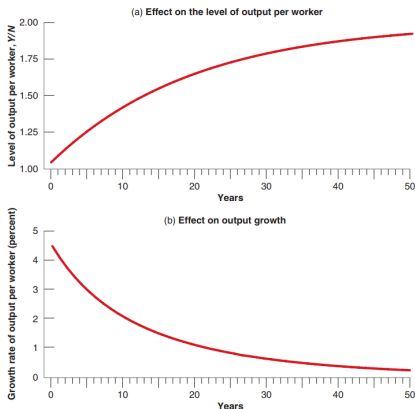
$$\begin{aligned} k_1 - 1 &= 0.2\sqrt{1} - (0.1)1 \\ k_1 &= 1.1 \end{aligned}$$

- In the same way, we can solve for k_2 , and so on.

Speed of Adjustment

11.3 Getting a Sense of Magnitudes

- Once we have determined the values of capital per worker in year 0, year 1, and so on, then we can use equation (11.6) to solve for output per worker in year 0, year 1, and so on.



The Golden Rule Level of Consumption

11.3 Getting a Sense of Magnitudes

- Recall that consumption is equal to what is left after saving.

$$C = Y - S = Y - sY$$

- Divide both sides by N and impose the Cobb-Douglas production function and steady-state condition:

$$c^* = \frac{C^*}{N} = \sqrt{k^*} - \delta k^*$$

$$c^* = \frac{s}{\delta} - \delta \left(\frac{s}{\delta} \right)^2$$

$$c^* = \frac{s(1-s)}{\delta}$$

- Steady-state consumption per worker is largest when s is equal to 0.5.

The Saving Rate and the Steady-State Levels

11.3 Getting a Sense of Magnitudes

- Note that the U.S saving rate since 1950 has only been about 17%.

Table 11-1 The Saving Rate and the Steady-State Levels of Capital, Output, and Consumption per Worker			
Saving Rate s	Capital per Worker K/N	Output per Worker Y/N	Consumption per Worker C/N
0.0	0.0	0.0	0.0
0.1	1.0	1.0	0.9
0.2	4.0	2.0	1.6
0.3	9.0	3.0	2.1
0.4	16.0	4.0	2.4
0.5	25.0	5.0	2.5
0.6	36.0	6.0	2.4
...	...	...	...
1.0	100.0	10.0	0.0

Extending the Production Function

11.4 Physical versus Human Capital

- We have concentrated so far on physical capital.
- But economies have another type of capital: the skills of the workers in the economy, or what economists call **human capital**.
- The most natural way to extending our analysis to allow for human capital is to modify the production function relation (11.1) to read

$$y = \frac{Y}{N} = f\left(\frac{K}{N}, \frac{H}{N}\right) = f(k, h) \quad (11.10)$$

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where H denotes human capital.

Growth: Endogenous, or Exogenous?

11.4 Physical versus Human Capital

- Models that generate steady growth *even without* technological process are called **models of endogenous growth**.
- Following the lead of Robert Lucas and Paul Romer, researchers have explored the possibility that the joint accumulation of physical capital and human capital might actually be enough to sustain growth.
- If these economists are right, growth is endogenous, because it depends on variables potentially under the control of policymakers and individuals.
- In the Solow model, by contrast, growth is exogenous, because it depends on the rate of technological progress, which is taken as given.